A PROPER MORPHISM CARRYING AN AMPLE INVERTIBLE SHEAF IS LOCALLY PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-proper-ample-locally-projective is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 10616-10620 and proof span 10622-10628 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. Judged under the fleet reading of 2026-10-14 that H1 is a rejection class rather than a difficulty grade: this proof is not a bare citation, an explicit omission or a pure restatement, because it replaces an immersion by a closed immersion using properness. The substantive step is named in difficulty_reason.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- Over an affine base, a finite type morphism with an ample invertible sheaf admits an immersion into a projective space over that base, cited from the same chapter. Cited at proof lines 10622-10624. Produces, locally on the base, the immersion whose target has to be upgraded to a closed subscheme.
- The image of a proper scheme over the base is closed, cited from the same chapter. Cited at proof lines 10625-10627. Turns the immersion into a closed immersion, because the source is proper over the base.

Source reference. Original lemma 01VT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2458–2458. Same-chapter reference used in the retained author text at source line(s) 9570, 9635, 9654, 9662, 9804, 10625. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01W6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2471–2471. Same-chapter reference used in the retained author text at source line(s) 10104, 10627, 10794, 10905, 11283, 13351, 14146. It is supplied only as a navigation pointer so that the retained text compiles; the

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Lemma 0.1. *Let $f : X \to S$ be a proper morphism of schemes. If there exists an f-ample invertible sheaf on X, then f is locally projective.*

Proof. If there exists an f-ample invertible sheaf, then we can locally on S find an immersion $i : X \to \mathbf{P}^n_S$, see Lemma 01VT. Since $X \to S$ is proper the morphism i is a closed immersion, see Lemma 01W6. □